



University of Management & Technology
Department of Computer Science
School of Systems and Technology
Fall Semester 2025

Instructor's Name: Mr. Sved Hamed Raza **Subject:** Deep Learning and Neural Networks
Course Code CS4152 **Assignment #:** 01
Total Marks 25 **Due Date:** 15-11-2025
Due Date:

Question 01 (10+10+5=25 Marks)

Part A:

Consider the following dataset for predicting house price (in lakhs):

Obs	x_1 (Size, 100 sq ft)	x_2 (Rooms)	y (Price)
1	8	3	120
2	10	4	155
3	6	2	95
4	12	5	185

1. Write the linear regression hypothesis function:

$$\hat{y} = w_1x_1 + w_2x_2 + b$$

2. Write the design matrix X , parameter vector $\theta = [b, w_1, w_2]^T$, and output vector y .

3. Compute $X^T X$ and $X^T y$.

Do not compute the inverse. Just write the matrices.

4. Use Row (Gauss) Elimination to solve for parameters b , w_1 , and w_2 . Show elimination steps.

5. Apply Gradient Descent manually for 3 iterations using:

- initial values: $b = 0, w_1 = 0, w_2 = 0$**
- learning rate: $\alpha = 0.001$**

In each iteration compute:

- predictions
- gradients for b, w_1, w_2
- updated parameter values (round to 3 decimals)

6. Predict the price of a house with:

- $x_1 = 9$
 - $x_2 = 4$
- using both parameter sets (Row Elimination & Gradient Descent).

Part B: Logistic Regression (Binary Classification)

A bank wants to predict whether a customer will approve a loan.

Obs	x_1 (Annual Income in lakhs)	x_2 (Credit Score / 100)	y (Loan Approved?)
1	4	6	0
2	6	7	1
3	5	5	0
4	8	8	1

The logistic model:

$$\hat{y} = \sigma(w_1x_1 + w_2x_2 + b)$$

where $\sigma(z) = \frac{1}{1+e^{-z}}$

1. Write the logistic regression hypothesis function using w_1, w_2, b .
2. For the initial parameters $b = 0, w_1 = 0, w_2 = 0$, compute the predicted probability $\hat{y}$ for each observation.
3. Compute the gradient updates for one iteration of Gradient Descent:

$$w_j := w_j - \alpha \frac{\partial J}{\partial w_j}, b := b - \alpha \frac{\partial J}{\partial b}$$

Use $\alpha = 0.01$

Round results to 3 decimals.

- 4. After updating parameters, compute the new predicted probability for Observation 2 ($x_1 = 6, x_2 = 7$). Interpret in one sentence whether the model predicts loan approval.**

Part C:

(Write 2–3 sentences)

- 1. Explain why linear regression uses mean squared error, while logistic regression uses cross-entropy loss.**
- 2. Explain why logistic regression predictions are always between 0 and 1, while linear regression predictions may be any real number.**